



Review Test Submission: Assignment Week 8

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Course EBN 122 S2 2017

Test Assignment Week 8

Started 9/14/17 8:11 AM

Submitted 9/18/17 5:30 PM

Due Date 9/18/17 11:59 PM

Status Completed

Attempt Score 16.4 out of 20 points

Time Elapsed 105 hours, 18 minutes

Instructions Read over the document *ClickUP Assignments Instructions*.

Before sending an enquiring regarding ClickUP question to any-one please verify that your problem is valid ... i.e. if you suspect incorrect marking first check with a fellow student or two.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final answer correct to 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use singular)
- Use 'u' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G**) and tera(**T**); do not use centi, deci and deca

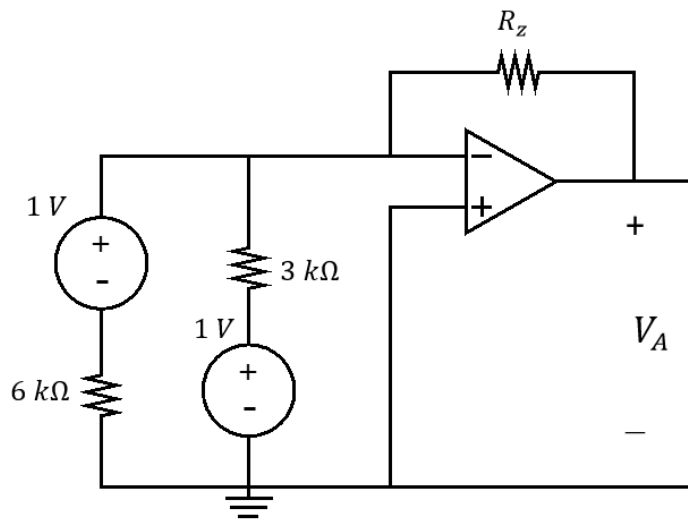
Results All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

2 out of 2 points



Given $R_Z = 16 \text{ k}\Omega$, determine V_A .



Selected Answer: ☒ -8V

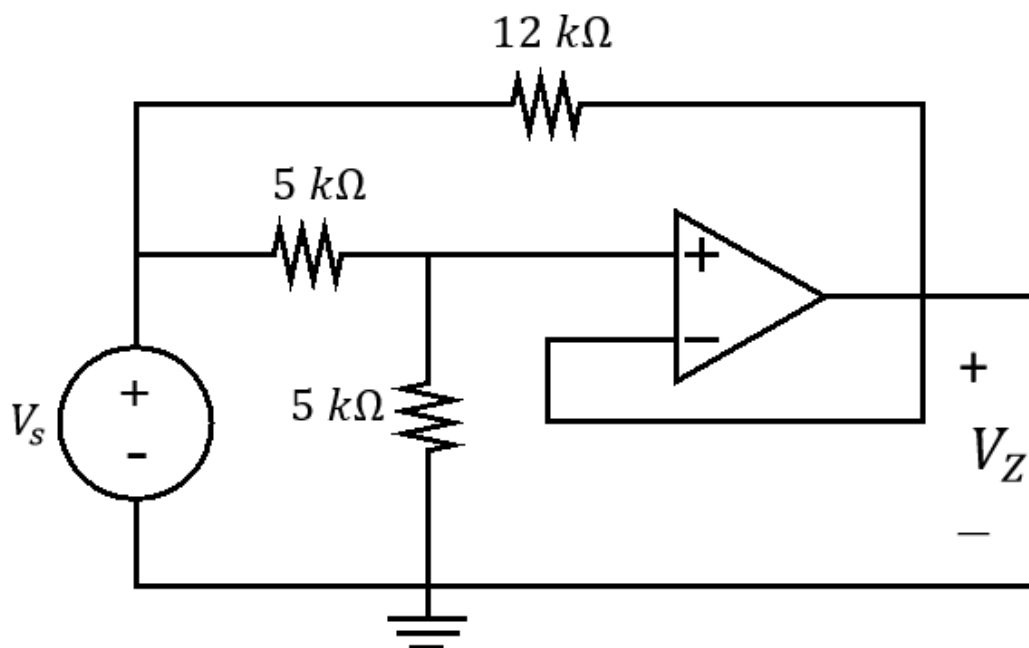
Correct Answer: ☒ -8.00 ± 5% (V)

Question 2

2 out of 2 points



Determine V_Z if $V_S = 17$



Selected Answer: ☒ 8.5V

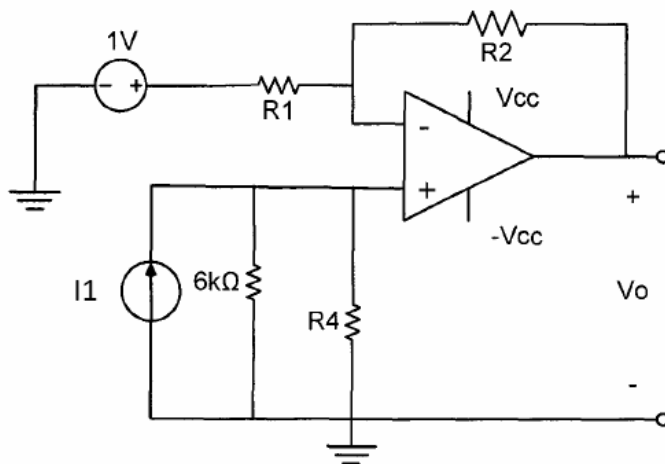
Correct Answer: ☒ 8.50 ± 5% (V)

Question 3

2 out of 2 points



If $R_1 = 10k\Omega$, $R_4 = 6k\Omega$ and $I_1 = 6\text{ mA}$, find R_2 so that $V_0 = 31\text{ V}$.



Selected Answer: ☒ 7.65kohm

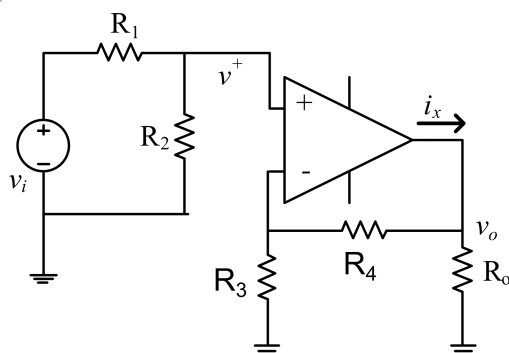
Correct Answer: ☒ 7.65 ± 5% (kohm)

Question 4

2 out of 2 points



Let $R_3 = 5 \text{ k}\Omega$ and $R_4 = 10 \text{ k}\Omega$. Find the current i_x when $R_o = 633 \Omega$ and $v^+ = -5 \text{ V}$.



Selected Answer: ☒ -24.7mA

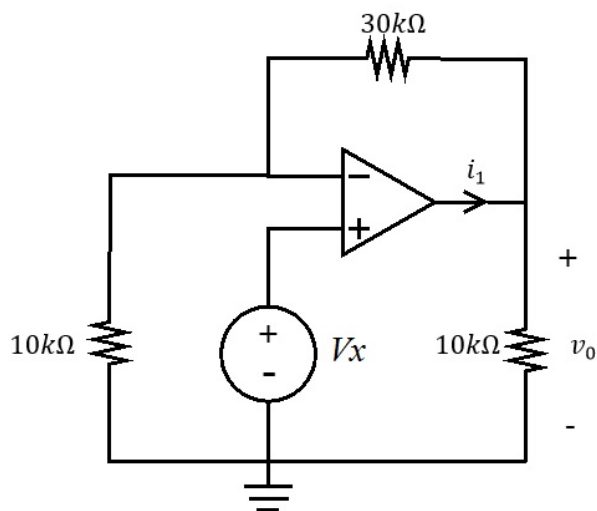
Correct Answer: ☒ -24.6967 ± 5% (mA)

Question 5

2 out of 2 points



If $V_x = 13 \text{ V}$ find V_o .



Selected Answer: ☒ 52V

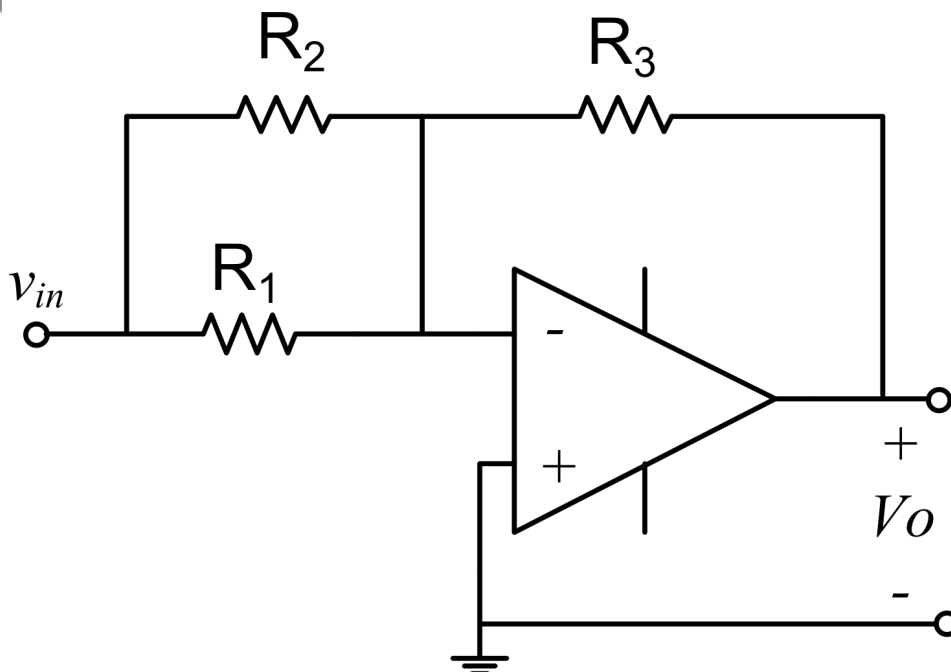
Correct Answer: ☒ 52.00 $\pm$ 5% (V)

Question 6

2 out of 2 points



Let $R_1 = 21\text{ k}\Omega$, $R_2 = 24\text{ k}\Omega$ and $R_3 = 76\text{ k}\Omega$. Calculate the output voltage V_o , if $v_{in} = 2\text{ mV}$



Selected Answer: ☒ -13.6mV

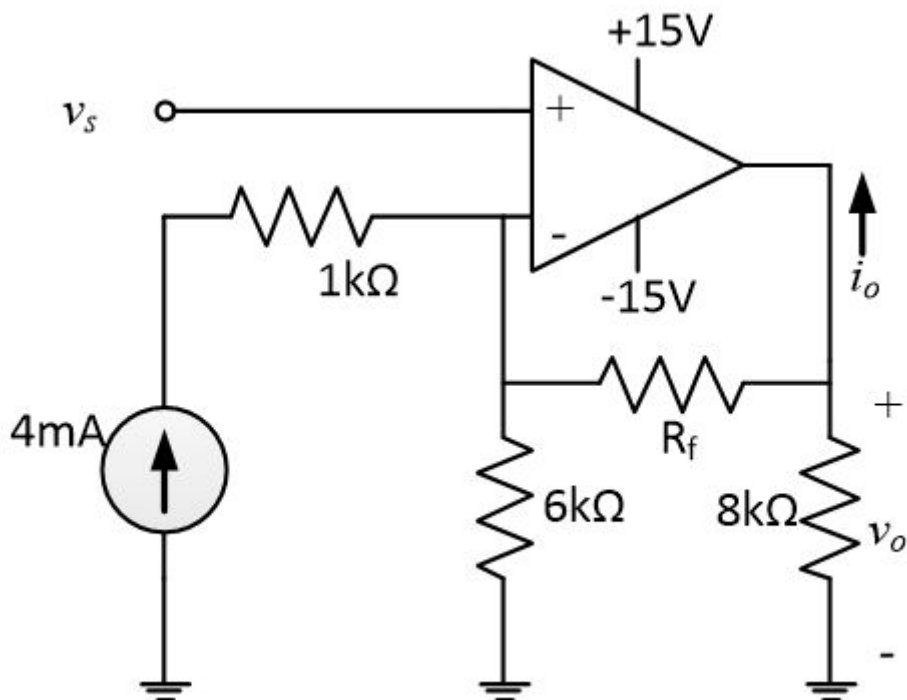
Correct Answer: ☒ -13.571 $\pm$ 5% (mV)

Question 7

0.2 out of 2 points



For what value of R_f (kOhms) will the opamp negatively saturate if $v_s = 1\text{ V}$.



Selected Answer: ✗ 1.83kohm

Correct Answer: ✓ $4.17 \pm 5\%$ (kohm)

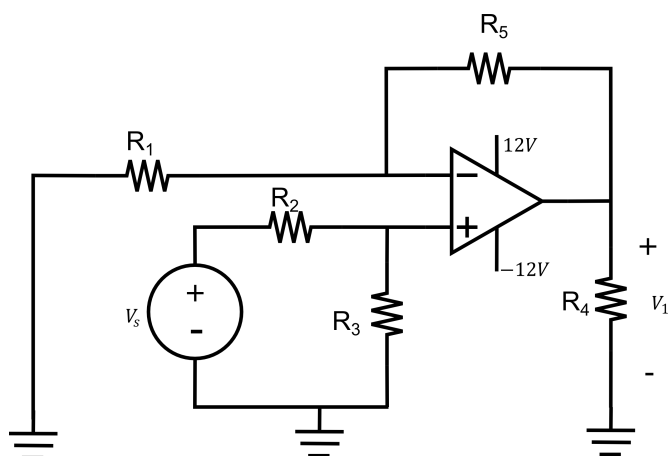
Question 8

2 out of 2 points



What is the maximum value V_s can assume to ensure that the operational amplifier operates in its linear range?

$R_1=7\Omega$, $R_2=28\Omega$, $R_3=12\Omega$, $R_4=7\Omega$ and $R_5=20\Omega$



Selected Answer: ✓ 10.4V

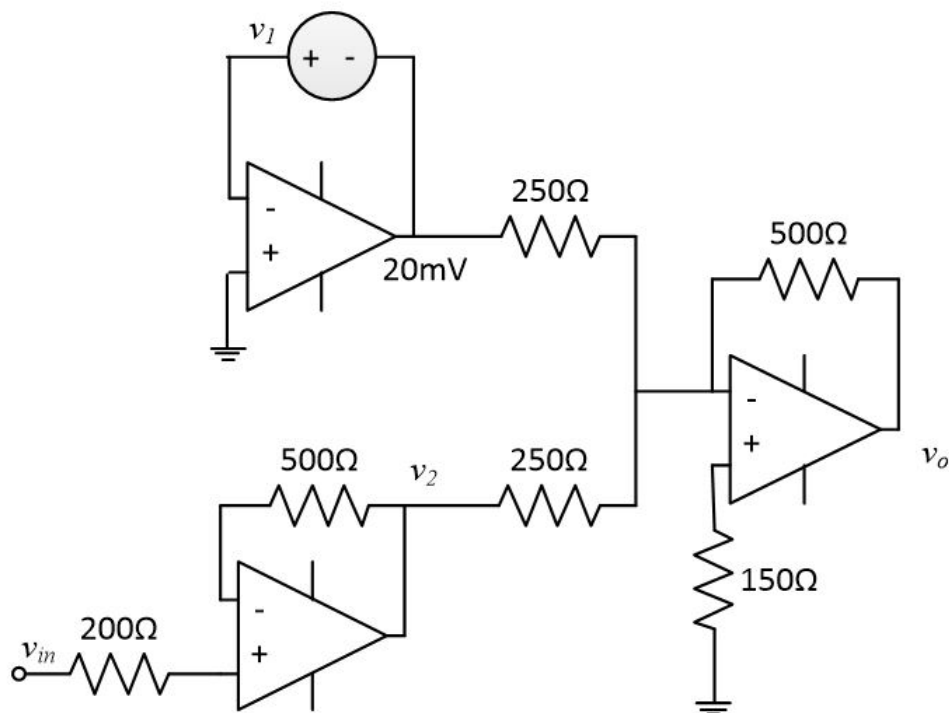
Correct Answer: ✓ $10.370 \pm 2\%$ (V)

Question 9

0.2 out of 2 points



Find v_o if $V_{in} = 2\text{ V}$



Selected Answer: ✗ -1.37V

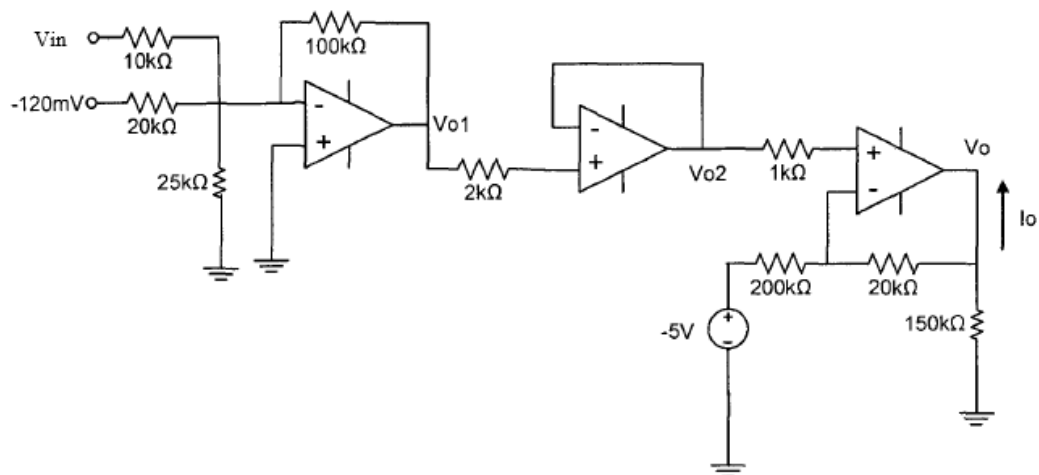
Correct Answer: ✓ $-4.04 \pm 5\%$ (V)

Question 10

2 out of 2 points



Assume $V_{o2} = 12.87$ V, determine V_o .



Selected Answer: ✓ 14.8V

Correct Answer: ✓ $14.66 \pm 5\%$ (V)

Monday, September 18, 2017 5:30:30 PM SAST

← OK

